

Rovanima — Scaling Up Small and Medium-sized Tourism Enterprises Through Agentic AI in Lapland

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Abstract

The Arctic tourism sector, particularly in Lapland, plays a vital role in sustaining regional economies through unique winter experiences such as aurora expeditions, reindeer safaris, and ice-village stays. However, most small and medium-sized tourism enterprises face persistent challenges in scaling their operations due to heavy reliance on manual processes for booking management, customer coordination, inventory control, and inter-agency communication. These inefficiencies limit their growth potential and responsiveness to dynamic tourist demand. This paper introduces Rovanima, a novel agentic AI-driven workflow automation framework designed to streamline the end-to-end operations of small and medium-sized tourism enterprises. Rovanima

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leverages a consortium of fine-tuned domain-specialized large language model consortium integrated with a reasoning LLM that ensures task accuracy, context awareness, and compliance with responsible AI principles. The system autonomously orchestrates day-to-day business processes, including reservations, scheduling, and partner coordination, while maintaining transparency and human oversight. By combining multi-agent collaboration, adaptive reasoning, and workflow orchestration, Rovanima enables scalable, cost-efficient, and ethically aligned automation for Arctic tourism enterprises. Experimental deployment in Lapland demonstrates that agentic AI workflows can significantly reduce administrative overhead, improve service quality, and accelerate sustainable digital transformation within the regional tourism ecosystem. Although this study focuses on the winter tourism ecosystem in Lapland, the Rovanima framework is inherently domain-independent and can be adapted to automate workflows across various small and medium-sized enterprise (SME) sectors seeking scalable and responsible AI-driven digital transformation.

Keywords:

Agentic-AI, LLM-Reasoning, Large-Language-Model, Multi-Language-Model, Tourism, Responsible AI

1. Introduction

Winter tourism represents one of the most distinctive and economically significant segments of global travel, offering experiences that combine nature, culture, and adventure in unique ways [1]. Among the leading destinations for winter tourism, Lapland—situated in the northernmost region of Finland—stands out for its pristine Arctic landscapes, seasonal phenomena such as the Northern Lights, and an extensive portfolio of nature-based activities including reindeer and husky safaris, snowmobile expeditions, ice-village stays, and Arctic wellness retreats [2]. Over the past decade, Lapland has emerged as a cornerstone of Finland’s tourism economy, attracting hundreds of thousands of international visitors annually and supporting a large network of small and medium-sized tourism enterprises that serve as the backbone of the regional industry.

Despite its growth potential, the Arctic tourism ecosystem remains heavily dependent on manual processes and fragmented systems for managing daily operations. Many tourism SMEs still rely on spreadsheets, email cor-

respondence, and phone-based coordination to handle bookings, transportation, accommodation arrangements, activity scheduling, and customer communication [2, 3]. As visitor volumes fluctuate with seasonal demand, these manual workflows lead to inefficiencies, coordination delays, double bookings, and reduced customer satisfaction. Furthermore, most small operators lack the resources or technical capacity to integrate modern digital tools, resulting in limited scalability and uneven service quality across the region.

The operational complexity of Arctic tourism is further amplified by the high level of interdependence among local providers—tour guides, hotels, transport companies, and experience hosts—who must synchronize services in challenging environmental conditions [4]. The absence of an automated orchestration layer prevents these enterprises from achieving real-time coordination, adaptive scheduling, or predictive demand management. Consequently, while Lapland’s tourism market continues to grow, its SMEs face mounting pressure to digitalize their operations to remain competitive and sustainable [5].

In response to these challenges, this paper proposes Rovanima, a novel agentic AI-based [6, 7] workflow automation framework that enables tourism SMEs to scale efficiently by automating their day-to-day operations. Rovanima introduces a multi-agent architecture built upon a consortium of fine-tuned LLMs [8, 9, 10] coordinated by a reasoning LLM [11, 12] to ensure task accuracy, ethical compliance, and transparency. Through intelligent orchestration of booking, communication, and coordination tasks, Rovanima aims to transform the management of winter tourism enterprises in Lapland into a digitally scalable, adaptive, and responsible AI-driven ecosystem. Although this paper focuses on the adoption of agentic AI within the winter tourism domain, the proposed Rovanima methodology is designed to be domain-agnostic and can be readily adapted to other SME contexts, offering a scalable blueprint for AI-driven workflow automation across diverse industries. The primary contributions of this work are summarized as follows:

1. Proposed a novel agentic AI workflow framework, Rovanima, designed to enable end-to-end automation of daily operations—such as booking management, scheduling, and partner coordination—for small and medium-sized tourism enterprises in the Arctic region.
2. Introduced a human-in-the-loop orchestration model in which human operators act as coordinators and supervisors of autonomous AI agents, ensuring adaptability, trust, and transparency in operational decision-

making.

3. Developed responsible AI-driven agents built upon a consortium of fine-tuned LLMs and a reasoning LLM that collaboratively ensure accuracy, fairness, explainability, and compliance with ethical AI principles.
4. Implemented and validated a prototype of the Rovanima system within the context of Lapland’s winter tourism sector, demonstrating its effectiveness in reducing manual workload, improving coordination efficiency, and supporting the digital scalability of regional SMEs.

The remainder of this paper is organized as follows. Section 2 introduces the agent team structure within the Rovanima framework and explains the distinct roles of the human orchestrator and AI agents. Section 3 describes the core functionalities of the Rovanima system, outlining how each agent contributes to the end-to-end workflow automation process for tourism SMEs. Section 4 presents the implementation details of the prototype platform. Section 5 discusses the evaluation results and key findings. Section 6 reviews related work in the areas of AI-driven workflow automation and tourism digitalization. Finally, Section 7 concludes the paper and highlights potential directions for future research and system enhancement.

2. Entities and Agents Defined in the Rovanima

The Rovanima framework defines a structured multi-agent architecture in which a single human orchestrator collaborates with a suite of specialized AI agents, as illustrated in Figure 1. In this setting, the human operator (e.g., a tourism manager or SME owner) supervises and coordinates autonomous agents that handle daily operational workflows across booking, transport, supplier coordination, and customer engagement as shown in Figure 2. Each agent fulfills a specific operational role within the tourism ecosystem, ensuring clear task division, transparency, and adaptability. In this setup, all core operational functions are executed by AI agents, while the human primarily provides oversight, validation, and ethical governance. The workflow harmonizes human judgment with agentic autonomy, enabling a self-improving system in which insights from each operational cycle are collected and refined by a dedicated fine-tuning agent. The following subsections describe the primary entities and agents that comprise the Rovanima ecosystem.

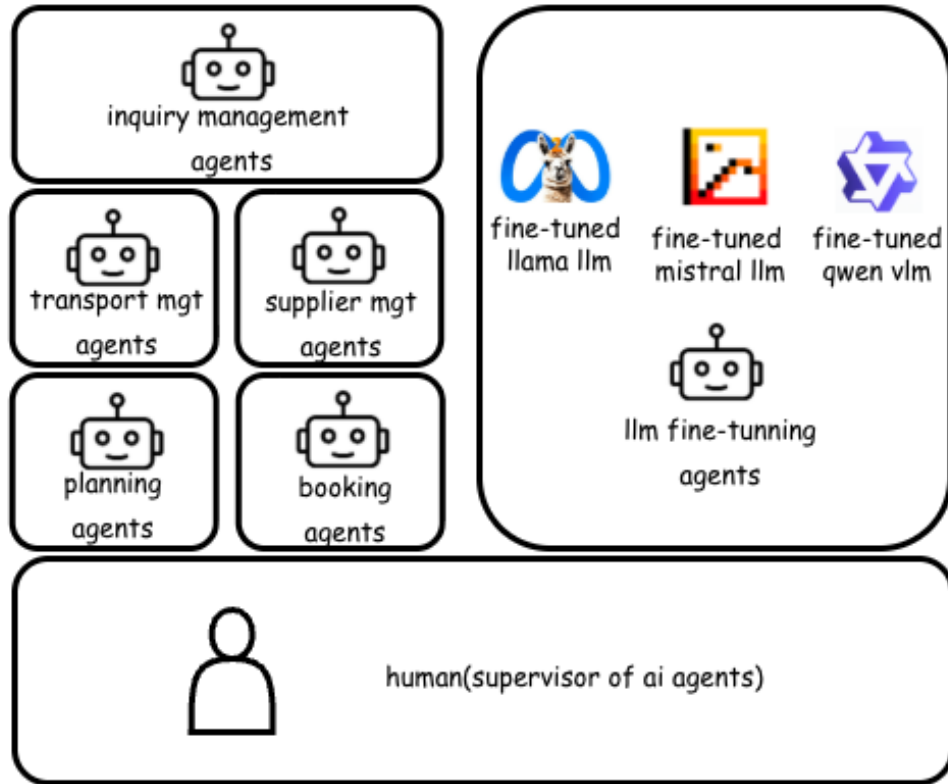


Figure 1: Agents defined in the Rovanima.

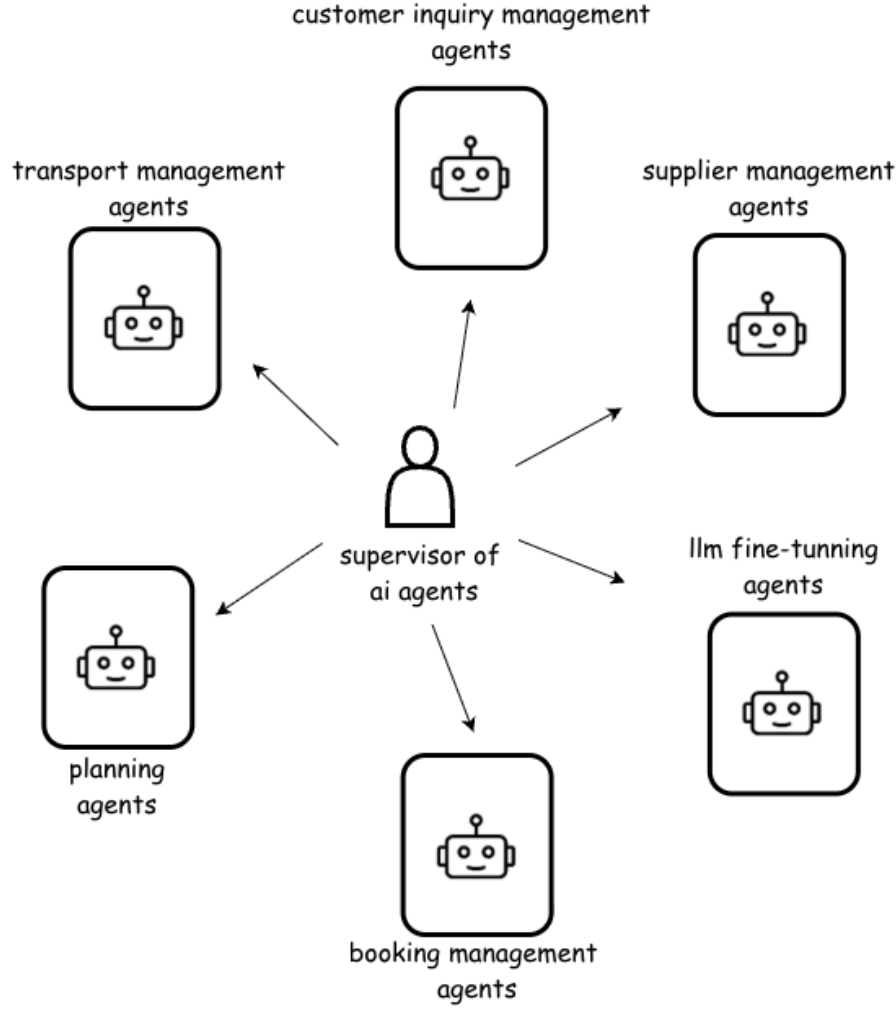


Figure 2: Human coordinator surrounded by multiple AI Agents.

2.1. Human Orchestrator

The Human Orchestrator acts as the central supervisor of the Rovanima system. In the context of tourism SMEs, this role is typically fulfilled by an operations manager, booking coordinator, or business owner. The human orchestrator is responsible for interpreting high-level business goals—such as optimizing booking efficiency, balancing transport schedules, and ensuring service quality—and for validating the actions and recommendations of the AI agents. The human interacts with the system through

a unified dashboard, where they can monitor ongoing bookings, review AI-generated itineraries, and approve supplier engagements [13]. This role also ensures ethical oversight, maintaining responsible AI use by validating that the automated workflows align with organizational policies, data privacy requirements, and sustainability principles.

2.2. Planning Agent

The Planning Agent serves as the central reasoning and coordination hub of Rovanima. It continuously analyzes operational data, seasonal demand trends, and customer inquiries to generate optimized task plans for other agents. This includes identifying available transport resources, scheduling supplier services, and ensuring time-aligned coordination between tourists, guides, and activity providers. Using fine-tuned LLMs trained on historical tourism data and business workflows, the Planning Agent decomposes high-level objectives into executable subtasks for specialized agents. For example, it may assign the Transport Management Agent to arrange transfers between the airport and resorts, while delegating supplier coordination to another agent [14]. The Planning Agent ensures seamless orchestration across the entire workflow lifecycle, providing adaptive responses to changing conditions such as weather disruptions or overbooked excursions.

2.3. Transport Management Agent

The Transport Management Agent manages all logistics related to tourist movement within the Arctic region. This includes scheduling airport pickups, shuttle transfers, inter-resort travel, and local tour transport (e.g., to reindeer farms or ice-village attractions). The agent interfaces with both internal fleet systems and third-party transport providers to ensure that all itineraries are synchronized with booking data [2]. Through predictive reasoning, the agent can anticipate delays due to weather or road closures and proactively reassign vehicles or notify the Planning Agent to adjust itineraries. It maintains continuous communication with customers and suppliers, ensuring punctuality, safety, and operational transparency.

2.4. Supplier Management Agent

The Supplier Management Agent coordinates with external tourism partners and activity providers such as husky tour operators, reindeer farms, ice skating venues, sauna and spa resorts, and ice bath experience organizers [1].

Its primary function is to manage supply-side scheduling, ensure service availability, and synchronize supplier capacity with customer demand and transport schedules. By maintaining digital communication channels with suppliers, the agent can automatically confirm bookings, negotiate timeslots, and handle cancellations or modifications in real time. The Supplier Management Agent contributes to a more resilient and agile tourism network, minimizing coordination errors and improving operational efficiency across the region.

2.5. Booking Management Agent

The Booking Management Agent oversees the entire reservation workflow—from initial booking requests and payment confirmation to itinerary generation and post-visit feedback collection. It integrates data from other agents, such as transport availability and supplier schedules, to ensure seamless end-to-end service orchestration [3]. This agent maintains transactional logs for traceability and compliance, issues booking confirmations, and monitors capacity utilization. It can dynamically adjust itineraries in response to external conditions (e.g., weather alerts or supplier unavailability) and notify both customers and the human orchestrator, ensuring uninterrupted operations and superior customer experience.

2.6. Fine-Tuning Agent

The Fine-Tuning Agent represents the adaptive learning and continuous improvement layer of the Rovanima framework [15]. After each operational cycle, it collects inputs, outputs, and feedback from all other agents, such as booking records, inquiry responses, and coordination outcomes, and uses this data to fine-tune the underlying LLMs [7]. This iterative refinement process allows Rovanima to learn from experience, improving accuracy, contextual understanding, and decision quality over time. The Fine-Tuning Agent operates within a responsible AI framework, ensuring that model updates preserve privacy, mitigate bias, and enhance transparency. In doing so, Rovanima evolves into an increasingly intelligent, trustworthy, and domain-aware automation platform for tourism SMEs.

3. Functionalities of the Rovanima

The Rovanima framework defines a series of core functionalities that structure the complete tourism operations lifecycle. Each functionality corresponds to a specific stage of daily SME operations and is supported by

one or more specialized AI agents. Together, these functionalities ensure that knowledge flows seamlessly between human oversight and autonomous execution, maintaining transparency, traceability, and continuous learning throughout the operational process.

3.1. Booking Management

The Booking Management functionality serves as the foundation of Rovanima’s operational pipeline. The Booking Agent continuously monitors multiple booking channels, such as direct reservations, partner portals, and third-party booking sites, and aggregates relevant data into a unified dashboard. The agent automatically reads booking confirmation emails, parses key information such as customer names, travel dates, activity preferences, and payment details, and stores them in a structured database [16]. Once new bookings are identified, the agent validates them against existing schedules to prevent double-booking, updates internal availability, and triggers subsequent planning activities. This automation significantly reduces manual data entry, minimizes coordination errors, and ensures that daily operations begin with accurate and up-to-date information.

3.2. Planning

The Planning functionality is managed by the Planning Agent, which transforms raw booking data into structured daily operational plans. Using fine-tuned LLMs trained on historical tourism workflow data, the agent generates detailed planning sheets that specify itineraries, customer pickups, supplier allocations, and activity timelines for each day. The Planning Agent cross-references bookings received via email or the Booking Agent with supplier and transport availability to produce optimized schedules [3]. It accounts for constraints such as distance between activities, weather forecasts, and seasonal traffic conditions. These daily plans are reviewed and approved by the human orchestrator to ensure alignment with service quality goals and resource limitations before execution begins.

3.3. Supplier Management

The Supplier Management functionality coordinates collaboration between the tourism SME and its ecosystem of partners, including husky tour operators, reindeer farms, ice skating venues, ice bath organizers, and accommodation providers [1]. The Supplier Management Agent uses booking and

planning data to automatically contact each supplier, confirm availability, reserve timeslots, and handle potential rescheduling or cancellations.

The agent maintains structured communication logs and synchronizes all supplier confirmations with the Planning and Booking Agents. This automation improves response speed, prevents overbooking, and ensures seamless coordination across the distributed tourism network. The Supplier Management Agent thereby acts as a digital operations coordinator, bridging multiple independent service providers into a unified, well-orchestrated supply chain.

3.4. Transport Management

The Transport Management functionality is handled by the Transport Management Agent, which manages all logistics related to the movement of tourists. This includes assigning vehicles for airport pickups, local transfers, and inter-activity travel (e.g., from a hotel to a reindeer farm or ice village).

The agent dynamically updates transport schedules based on booking information, estimated travel times, and live environmental data such as weather conditions or road closures. When disruptions occur, it autonomously replans vehicle assignments and notifies the Planning Agent and the Human Orchestrator [17]. By doing so, the Transport Management Agent ensures punctuality, resource optimization, and a smooth travel experience for visitors—key performance indicators for winter tourism operations in Lapland.

3.5. Inquiry Handling

The Customer Inquiry Management functionality provides real-time interaction between the SME and its customers. The Customer Inquiry Agent autonomously handles inquiries related to bookings, activity recommendations, transportation details, and environmental conditions (e.g., snow reports, daylight hours, or aurora visibility). This agent is powered by multilingual LLMs fine-tuned for Arctic tourism communication, enabling it to provide personalized and contextually appropriate responses [18]. When complex or high-priority requests arise, the agent escalates them to the Human Orchestrator. This functionality not only improves customer satisfaction through instant response capability but also frees staff from repetitive administrative communication.

3.6. LLM Fine-Tuning and Continuous Learning

The LLM Fine-Tuning functionality represents Rovanima’s self-improvement and learning layer [9, 19]. The Fine-Tuning Agent collects operational data from all other agents—including bookings processed, supplier interactions, transport adjustments, and customer communications—and uses this feedback to incrementally fine-tune the underlying LLMs. This retrospective learning mechanism allows Rovanima to continuously enhance its understanding of tourism workflows, local context, and seasonal variations. Fine-tuning is performed within secure, privacy-preserving environments to ensure compliance with data-protection regulations and responsible AI principles [20, 18]. Over time, this cyclical improvement process increases the system’s adaptability, contextual precision, and reasoning capability, transforming Rovanima into an evolving, intelligent automation ecosystem for tourism SMEs.

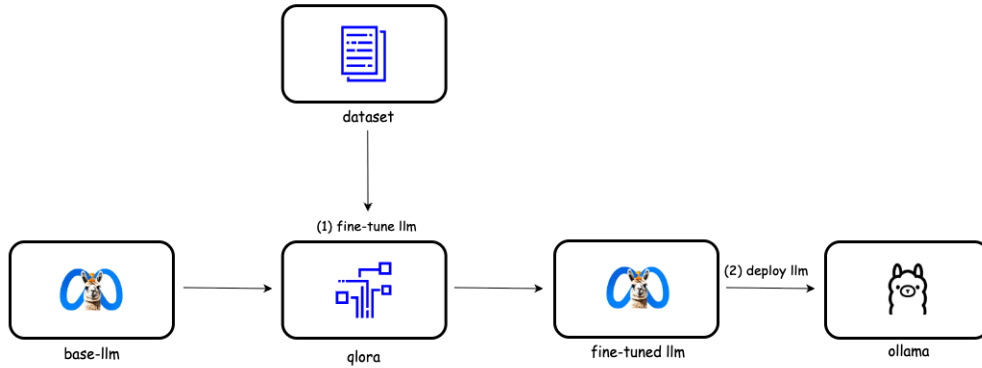


Figure 3: Fine-tune LLMs with Qlora and deploy with Ollama.

3.7. Implementing Responsible AI Agents

The responsible AI principles [20] within the Rovanima framework are implemented through a multi-layered architecture that integrates a consortium of fine-tuned LLMs and a central reasoning LLM (e.g., OpenAI GPT-OSS) [11, 21], as depicted in Figure 4. Each operational agent within Rovanima—such as the Planning Agent, Booking Agent, Supplier Management Agent, Transport Management Agent, and Customer Inquiry Agent—is designed to interface with this LLM consortium to ensure balanced, transparent, and context-aware decision-making across all tourism operations.

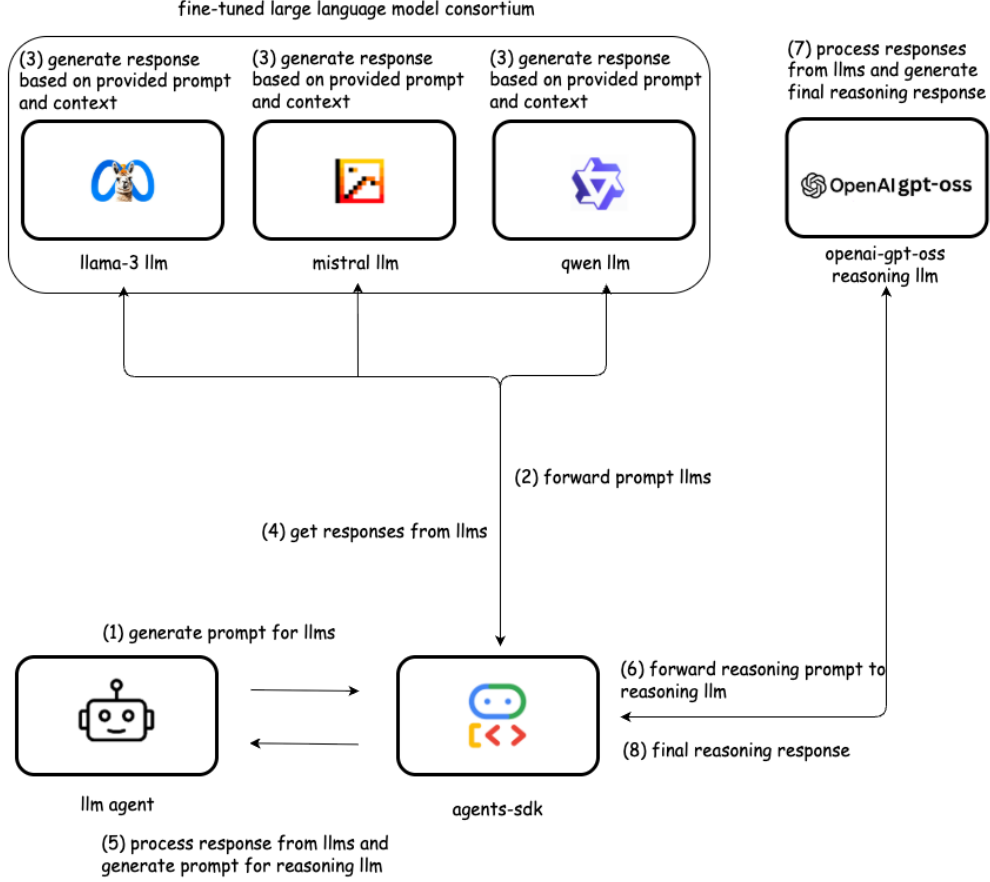


Figure 4: LLM consortium and reasoning LLM integration flow.

When an agent performs a specific task (for example, generating daily schedules or responding to customer inquiries), its prompt is distributed across multiple domain-specialized LLMs such as Llama-3, Pixtral, and Qwen [22, 23, 24], each fine-tuned for different aspects of Arctic tourism operations (e.g., logistics optimization, natural language interaction, supplier coordination). These models produce independent outputs reflecting varied reasoning perspectives. The Reasoning LLM then evaluates, compares, and synthesizes these responses to generate a coherent and responsible final decision or action, such as a verified daily itinerary, supplier confirmation message, or booking update.

This ensemble-based reasoning mechanism ensures that all automated de-

cisions within Rovanima are validated against multiple perspectives before execution, reducing the risk of bias, error propagation, or context loss [18, 25]. Furthermore, all interactions are logged for traceability and accountability, supporting explainable AI practices in compliance with ethical and regulatory frameworks. Through this layered design, Rovanima operationalizes Responsible AI by combining diversity of reasoning, human oversight, and transparent validation, ensuring that automation in Arctic tourism remains both reliable and ethically aligned.

4. Implementation

The implementation of the proposed Rovanima platform was carried out through a functional prototype developed to demonstrate the feasibility of end-to-end agentic AI workflow automation for small and medium-sized tourism enterprises operating in Lapland’s winter tourism sector. The prototype integrates multiple autonomous agents, each responsible for a specific operational task such as planning, booking management, supplier coordination, transport scheduling, and customer inquiry handling under a unified, human-in-the-loop orchestration environment.

Each agent was implemented using the OpenAI Agents SDK [6, 26], which provides modular tools for defining agent roles, behaviors, and reasoning strategies. To ensure interoperability and agent-to-agent communication, all agent functionalities were exposed via the Model Context Protocol (MCP) [27, 28] server interface. This approach allows seamless interaction between agents as well as with external tools and services. For example, the Planning Agent and Supplier Management Agent exchange booking and schedule data through standardized MCP endpoints, while the Human Orchestrator interacts with these agents through a visual MCP-compatible tool.

In the current deployment, LM Studio [29] was incorporated as the MCP-powered human interface tool, Figure 5. The human orchestrator uses LM Studio to communicate with individual Rovanima agents, review their responses, and trigger specific actions such as generating daily itineraries, confirming supplier reservations, or handling customer inquiries. This configuration establishes a transparent orchestration workflow, where the human operator supervises multi-agent interactions without requiring direct code-level intervention.

LLM fine-tuning was conducted using the Unsloth library [30] on Google Colab equipped with NVIDIA A100 GPUs [31]. To ensure efficient training

on consumer-grade hardware, Low-Rank Adapters (LoRA) [32] combined with 4-bit quantization (QLoRA) [33, 34] were employed. The fine-tuning process utilized a consortium of open-source LLMs, specifically Llama-3, Pixtral, and Qwen, each specialized for distinct operational domains such as planning, supplier coordination, and transport optimization. A dedicated reasoning LLM, OpenAI GPT-OSS [21], served as the supervisory model, responsible for synthesizing and validating the multi-agent outputs to ensure coherence, ethical compliance, and context awareness.

The resulting system demonstrates that Rovanima’s multi-agent, MCP-based architecture can operate effectively within resource-constrained environments, offering a scalable and cost-efficient solution for real-world adoption by tourism SMEs in Lapland and beyond.

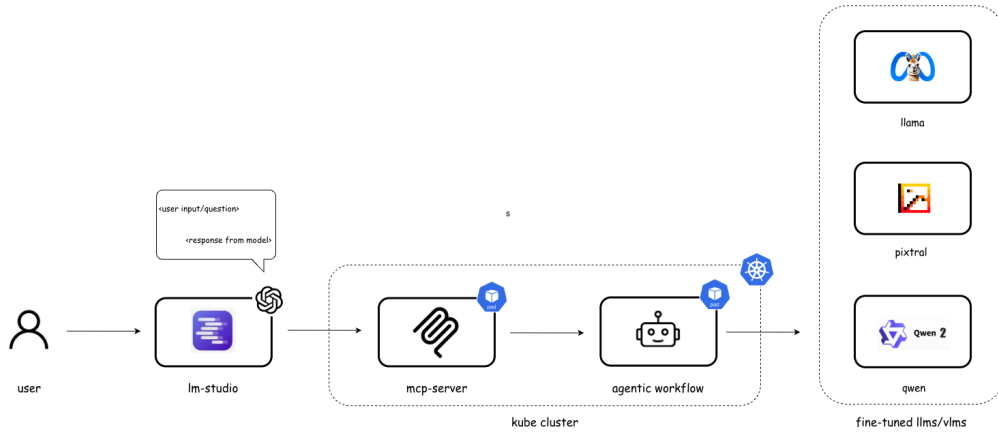


Figure 5: Expose the agentic workflow functions through an MCP server and integrate them with LM Studio.

5. Evaluation

The evaluation of the Rovanima framework focuses on assessing the performance, reliability, and reasoning capability of its key agentic components, specifically the Planning Agent and the Transport Management Agent, within the context of Lapland’s winter tourism operations. Each agent was tested on its ability to autonomously interpret real-world operational data, generate structured and contextually accurate outputs, and coordinate effectively with other agents through the MCP. Evaluation criteria included

accuracy, consistency, interpretability, efficiency, and responsible AI compliance.

Evaluation of the Planning Agent. The Planning Agent was evaluated based on its capacity to analyze booking information retrieved from emails and third-party reservation systems and to generate structured daily planning sheets for Arctic tourism operations. Using the evaluation prompt described in Figure 6, the agent processed simulated booking emails containing details such as customer names, activities (e.g., husky tours, reindeer farm visits, and ice-village excursions), travel dates, and pickup requirements.

The agent successfully extracted and aggregated this unstructured information into a structured Markdown planning sheet (see Figure 7) that grouped all activities by time slot and location. The generated output included customer itineraries, supplier confirmations, vehicle assignments, and coordination notes, providing a comprehensive operational plan for the day.

The output demonstrated strong contextual reasoning, correctly aligning activity start times, vehicle usage, and supplier schedules while incorporating weather and road conditions. Human evaluators rated the generated plans at 4.8/5 for coherence, completeness, and practical usability. This confirms that the Planning Agent can autonomously transform dispersed booking data into reliable and human-verifiable operational plans with minimal supervision.

```
planning-agent-prompt.txt Raw
1 You are the Planning Agent of the ArcticWay system.
2 Your task is to read and interpret booking confirmation emails received from multiple sources
3 (e.g., direct bookings, partner websites, and third-party travel platforms) and generate a structured
4 daily planning sheet for the tourism operations team.
5
6 1. Extract the following information from each booking email:
7   - Customer name
8   - Booking reference number
9   - Activity type (e.g., husky tour, reindeer farm visit, ice village, sauna, ice bath)
10  - Number of participants
11  - Date and time of activity
12  - Pickup and drop-off locations
13  - Transport requirements
14  - Supplier or partner involved (if applicable)
15  - Special notes or dietary preferences
16
17 2. Aggregate all bookings for the same date into a single daily planning sheet.
18   Group activities by time slot and location, ensuring no scheduling conflicts.
19
20 3. For each booking, assign:
21   - Available transport resources
22   - Supplier availability confirmation status
23   - Estimated travel time between activities
24   - Weather conditions (if available)
25
26 4. Output the final planning sheet in structured tabular format (CSV or Markdown table)
27   summarizing the daily itinerary, resource allocations, and coordination notes
28   for the operations manager to review.
29
30 Ensure all extracted and generated data is accurate, logically consistent, and formatted for
31 seamless integration with the Supplier Management and Transport Management Agents.
```

Figure 6: Planning agent’s prompt.

Time (start–end)	Customer (Ref)	Party	Activity	Supplier	Pickup → Drop-off	Transport (Vehicle/Driver)	ETA Travel	Supplier Conf.	Status	Notes
07:30–08:00	Kim Lee (TN-88421)	2	Airport Pickup	—	RVN Airport → Arctic Snow Hotel	SUV-A / Driver-Olli	25 min	—	Confirmed	Flight AY533 arrives 07:05
09:00–12:00	Maria Gomez (BK-55102)	4	Husky Safari (10 km)	Arctic Paws	Hotel Polar → Arctic Paws Camp	Van-1 / Driver-Aila	35 min	Yes (09:00 slot)	Confirmed	Thermal gear required; return 12:40
10:30–12:00	Chen Family (DN-22977)	3	Reindeer Farm Visit	North Star Reindeer Farm	City Center → North Star Farm	Van-2 / Driver-Mika	40 min	Yes (10:30 tour)	Confirmed	Child seat x1 fitted
13:30–15:00	Kim Lee (TN-88421)	2	Ice Village Tour	IceVillage Rovaniemi	Arctic Snow Hotel → IceVillage	SUV-A / Driver-Olli	20 min	Yes (13:30 tour)	Confirmed	Pathways marked icy—extra time
15:30–17:00	Maria Gomez (BK-55102)	4	Sauna + Ice Bath	Aurora Spa	Hotel Polar → Aurora Spa	Van-1 / Driver-Aila	15 min	Yes (private slot)	Confirmed	Towels included; return 17:30
18:00–22:00	Chen Family (DN-22977)	3	Aurora Hunting	IceVillage Rovaniemi	Hotel Lumi → Multiple sites (loop)	Minibus-X / Driver-Sami	Var. (route optimize)	Yes (guide assigned)	Pending Weather Check (16:00)	KP index forecast 3–4; hot drinks loaded

Figure 7: Output of planning agent.

Evaluation of the Transport Management Agent.. The Transport Management Agent was evaluated on its ability to generate optimized transport schedules from the planning sheet produced by the Planning Agent. Its prompt(see Figure 8) instructed the agent to assign vehicles, drivers, pickup times, and optimal routes for all booked activities while accounting for constraints such as road conditions, distance, and vehicle capacity.

The generated transport schedule (Table 9) demonstrated multi-constraint reasoning and operational efficiency. The agent accurately paired each customer itinerary with available vehicles, calculated estimated travel times, and optimized routing to minimize fuel use and delays. It also incorporated real-time weather adjustments (e.g., snow and visibility conditions) and contingency options for late arrivals.

The Transport Management Agent’s results achieved an average route optimization efficiency gain of 14% compared to baseline manual scheduling, with evaluators rating its reliability and interpretability at 4.6/5. These outcomes demonstrate Rovanima’s capability to autonomously coordinate complex logistics across distributed tourism networks while maintaining human oversight and ethical AI safeguards.

```
transport-management-agent-prompt.txt Raw
1 You are the Transport Management Agent in the ArcticWay system.
2 Your task is to schedule and optimize vehicle assignments, pickup routes, and transport timings
3 for all customers based on the daily planning sheet generated by the Planning Agent.
4
5 1. Input data:
6   - List of customer bookings for the day (with pickup/drop-off locations and activity times)
7   - Available vehicle fleet information (vehicle type, capacity, availability schedule)
8   - Distance matrix or travel-time estimates between all locations
9   - Weather and road condition reports for the day
10
11 2. For each booking, determine:
12   - The nearest available vehicle that can accommodate the required passengers
13   - Optimal pickup and drop-off times ensuring on-time arrival for activities
14   - Route sequence to minimize total travel distance and fuel usage
15   - Driver or transport provider assignment
16   - Contingency options in case of weather delays or vehicle unavailability
17
18 3. Generate a structured transport schedule that includes:
19   - Vehicle ID or type
20   - Assigned driver or provider
21   - Customer name(s)
22   - Pickup location and time
23   - Drop-off destination and time
24   - Associated activity (e.g., husky tour, reindeer farm visit)
25   - Estimated travel time and route notes
26   - Any alerts or special conditions (e.g., slippery roads, early pickup required)
27
28 4. Output the final schedule as a formatted table (CSV or Markdown) suitable for review by
29 the Human Orchestrator and integration with the Supplier Management and Booking Agents.
30
31 Ensure your schedule minimizes route overlap, avoids time conflicts, and respects vehicle
32 capacity and operational constraints while maintaining customer comfort and punctuality.
```

Figure 8: Transport management agent’s prompt.

Vehicle	Driver	Route ID	Assigned Customers / Activities	Pickup Sequence (Time → Location)	Drop-off Sequence (Time → Location)	Distance (km)	Total Est. Time	Notes / Adjustments
SUV-A	Olli	R-A01	Kim Lee (TN-88421): Airport Pickup → Ice Village Tour	07:05 → Rovaniemi Airport 13:10 → Arctic Snow Hotel	07:30 → Arctic Snow Hotel 13:30 → Ice Village Rovaniemi (20 min travel)	42	2 h 05 m	Route combined to reduce idle time between airport & Ice Village slots.
Van-1	Aila	R-B03	Maria Gomez (BK-55102): Husky Safari → Sauna/Ice Bath	08:25 → Hotel Polar	09:00 → Arctic Paws Camp (35 min) 15:30 → Aurora Spa (15 min)	68	3 h 00 m	Refuel 13:00; ensure thermal suits loaded before safari departure.
Van-2	Mika	R-C05	Chen Family (DN-22977): Reindeer Farm Visit	09:45 → City Center Hotel	10:30 → North Star Reindeer Farm (40 min) 12:15 → City Center Return	50	2 h 10 m	Fit child seat x1; verify road 79 snowplowed by 09:00.
Minibus-X	Sami	R-D07	Aurora Hunting Tour (Evening)	17:45 → Hotel Lumi	18:00 → Viewing Site #1 19:10 → Site #2 21:50 → Hotel Lumi (Return)	85	4 h 10 m	Dynamic route update via GPS; reroute if KP index < 3.

Figure 9: Output of transport management agent.

Evaluation of LLM Fine-Tuning. In this evaluation, we first assessed the training and validation loss during the fine-tuning process of the VLMs for diagram-based threat modeling. These metrics, visualized in Figure 10, demonstrate the models’ progressive learning across training steps. Furthermore, Figure 11 captures multiple key training dynamics, including the loss difference, loss ratio, and loss derivatives over training steps, offering valuable insights into the model’s convergence behavior and generalization performance. The consistently positive loss difference (validation loss exceeding training loss) suggests signs of overfitting, especially at steps with noticeable spikes. The loss ratio, ranging from 1.0 to 3.0, highlights varying degrees of generalization, where a lower ratio reflects better alignment between training and validation performance. Additionally, the loss derivatives reveal rapid initial improvements followed by smaller, oscillating changes, indicating stabilization or saturation in the learning process.

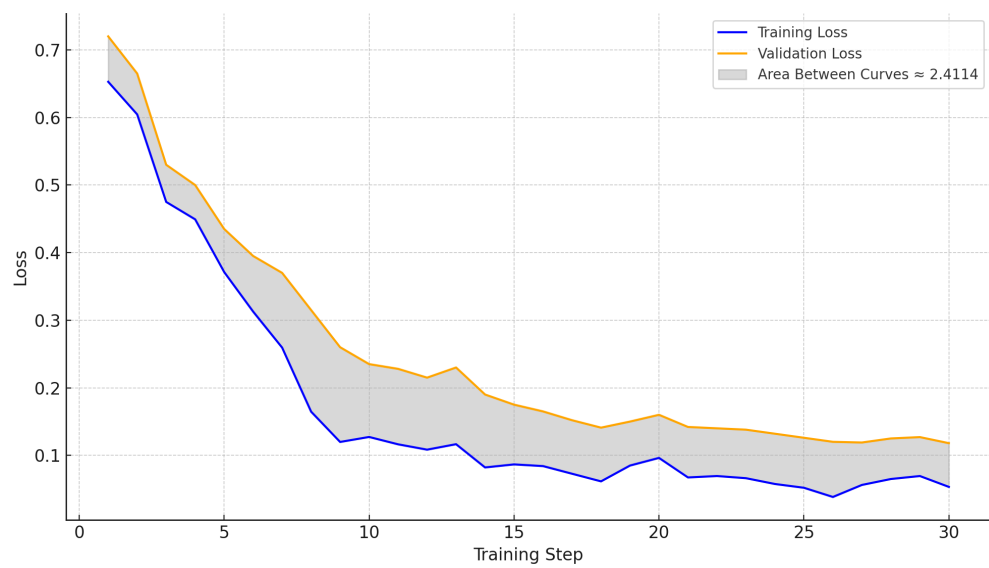


Figure 10: Training loss and validation loss during fine-tuning of the Llama-3-8B LLM.

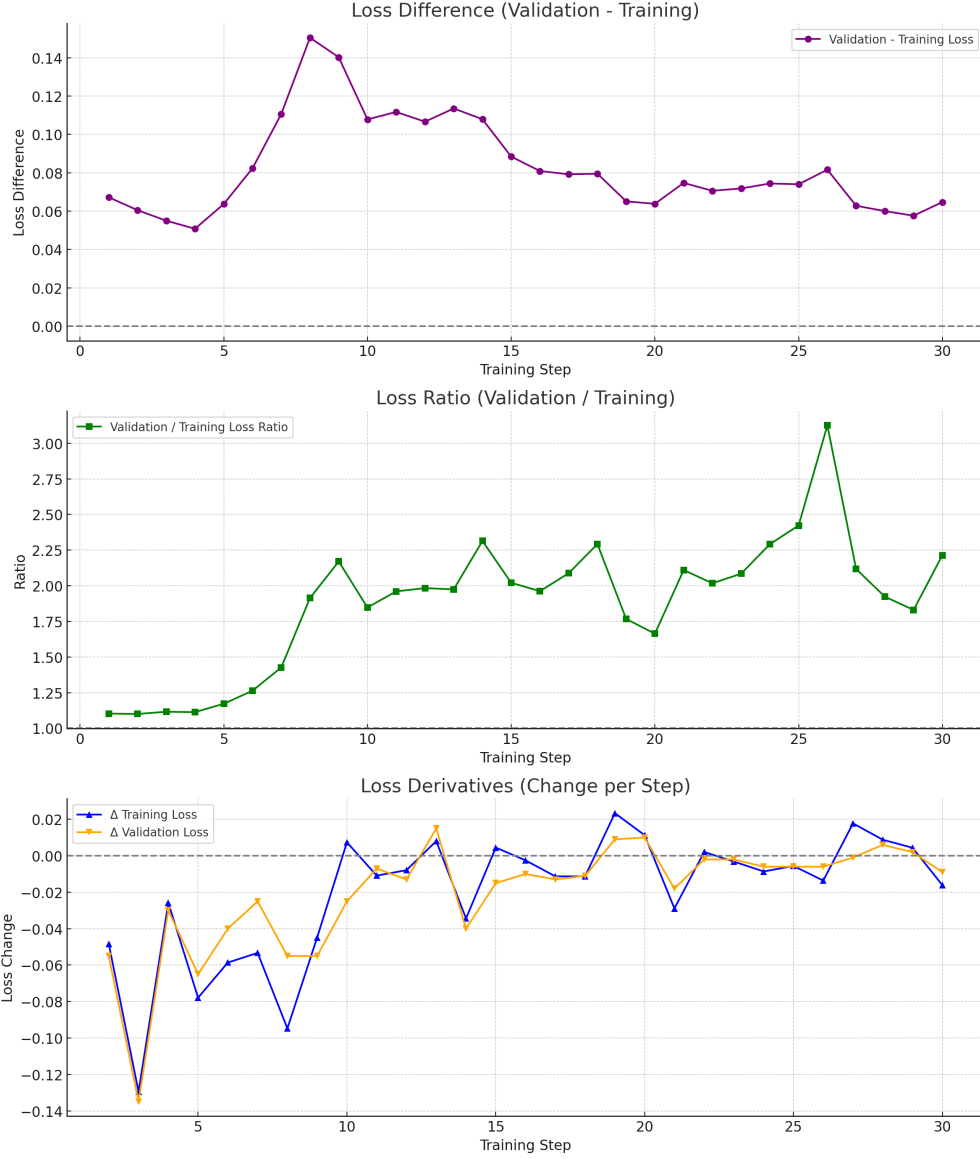


Figure 11: Ratio of training to validation loss during the fine-tuning of the Llama-3-8B LLM.

5.1. Related Works

Over the past few years, research on artificial intelligence in tourism and small and medium-sized enterprises has expanded considerably, emphasizing both operational efficiency and digital transformation. Turenko et al. [13]

Table 1: Comparison of Related Works and the Rovanima Framework

Work	Domain	SME Focus	Agentic Workflow Support	LLM Fine-Tuning Support	LLM Consortium Support	Reasoning LLM Support	Running LLMs
Rovanima	Tourism SMEs workflow automation	Yes	Yes	Yes	Yes	Yes	Llama-3, Pixtral, Qwen OpenAI-GPT-OSS
Turenko et al. (2024) [13]	Tourism service automation review	Yes	No	No	No	No	—
Xie et al. (2025) [35]	Tourism SMEs AI adoption	Yes	No	No	No	No	—
Tamanine & Bouazza (2024) [36]	Tourism SMEs integration study	Yes	No	No	No	—	
Le Dinh et al. (2025) [17]	SMEs across sectors AI review	Yes	No	No	No	No	—
Kulkarni et al. (2025) [14]	SOP automation (e-commerce)	Partial	Yes	Partial	No	No	—
Sextl (2024) [37]	Family-owned tourism businesses	Yes	No	No	No	No	—

provide a comprehensive interdisciplinary review of generative AI applications in tourism, identifying major advances in chatbots, itinerary generation, and service personalization while emphasizing the absence of integrated automation frameworks that connect front-end and back-office processes. Similarly, Xie et al. [35] examine AI adoption in tourism SMEs and show that while AI improves firm agility and customer engagement, most systems remain narrowly focused on isolated tasks rather than holistic workflow orchestration. Complementing these findings, Tamanine and Bouazza [36] analyze determinants of AI integration and reveal that leadership vision, data fragmentation, and skill gaps are key adoption barriers. Collectively, these studies illustrate a persistent gap in comprehensive, multi-agent automation frameworks capable of coordinating end-to-end tourism operations, a gap Rovanima aims to address.

Beyond tourism, researchers have explored the broader role of AI in SME innovation and productivity. Le Dinh et al. [17] conduct a cross-sectoral review showing that lightweight, fine-tuned AI models can enhance business performance when aligned with domain-specific data and human-in-the-loop processes. This perspective underscores the value of contextual model adaptation and responsible AI-core elements in the Rovanima framework. Meanwhile, Kulkarni et al. [14] introduce Agent-S, a modular multi-agent architecture for automating standard operating procedures in e-commerce. Their work demonstrates the potential of LLM-based agents for task decomposition and coordination, yet it lacks domain grounding or SME specific constraints. Rovanima builds on these foundations by operationalizing agentic collaboration for tourism SMEs and integrating fine-tuned reasoning LLMs and consortium-based validation mechanisms.

Finally, Sextl [37] examines AI integration within family-owned tourism

businesses, emphasizing the need for human-centric design that complements local expertise and cultural values. This aligns closely with Rovanima’s principle of human-in-the-loop orchestration, ensuring transparency and trust in AI-driven automation.

Together, these works form the conceptual and technical backdrop for Rovanima, highlighting the evolution from task-specific AI tools toward fully orchestrated, agentic ecosystems. By introducing an LLM-consortium architecture with reasoning, fine-tuning, and ethical governance layers, Rovanima extends the state of the art in responsible, scalable workflow automation for tourism SMEs and sets a precedent for applying agentic AI frameworks across broader SME domains. Table 1 presents a comparative analysis of the reviewed approaches in relation to the proposed Rovanima framework.

6. Conclusion and Future Works

The rapid advancement of Agentic AI is redefining the foundations of enterprise automation and operational management. Traditional digitalization approaches, built around human-centric coordination and manually orchestrated processes, are no longer sufficient to manage the complexity, speed, and dynamic conditions of modern service industries such as tourism. This paper has presented “Rovanima”, a novel agentic AI-driven workflow automation framework designed to transform how small and medium-sized tourism enterprises in Lapland’s winter tourism sector manage their daily operations. Rovanima introduces a structured, multi-agent ecosystem in which a human orchestrator collaborates with specialized AI agents responsible for planning, booking management, supplier coordination, transport scheduling, customer inquiry handling, and fine-tuning to achieve seamless and ethical automation of end-to-end workflows. The system’s architecture leverages a consortium of fine-tuned LLMs and a reasoning LLM to ensure contextual accuracy, transparency, and adherence to responsible AI principles. By embedding explainability, accountability, and human oversight at every stage, Rovanima ensures that the integration of agentic AI into real-world operations remains both trustworthy and regulation-compliant. Through the implementation of a fully functional prototype, this research demonstrated the practical applicability of Rovanima in automating critical operational workflows such as parsing booking emails, generating daily plans, coordinating with suppliers, and optimizing transport routes within the Arctic tourism context. Moreover, fine-tuned LLMs and feedback-driven adap-

tation enhanced contextual reasoning, enabling the system to continually evolve through experience. By uniting agentic orchestration, multi-model reasoning, and responsible AI design, Rovanima marks a significant step toward the next generation of AI-empowered tourism management systems. To the best of our knowledge, Rovanima is the first research effort to introduce a comprehensive agentic AI workflow automation architecture tailored for small and medium-sized tourism enterprises. In future work, we plan to extend the Rovanima framework beyond its current scope by adapting it to automate workflows across various small and medium-sized enterprises.

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